

MUHAMMAD ABDULLAHI

Room 010, Level 4, School of Mathematical Science Building 15 | City Campus. | +(61) 0452114730 | 4.msabdullahi@gmail.com
Google Scholar | ResearchGate | Web of Science

Education

- Royal Melbourne Institute of Technology, Victoria, Australia (RMIT) | PhD 2025-Till date
- Central South University, Changsha, PR China (CSU) | M.S Applied Mathematics 2020-2023
- Sule Lamido University Kafin Hausa, Nigeria (SLUK) | B.Sc Mathematics 2014-2018

Research Experience

- ✦ Research Associate | School of Mathematical Science, RMIT 2025-Till date
Key Contributions:

- Conduct advanced research in numerical optimization, nonsmooth and nonconvex optimization, and primal-dual splitting algorithms for solving large-scale optimization problems.
- Develop and analyse accelerated iterative algorithms, establishing convergence properties including global convergence, convergence rates, and computational complexity.
- Design and implement optimization algorithms using MATLAB and Python, and perform extensive numerical experiments on large-scale benchmark problems.
- Apply optimization techniques to real-world applications such as image restoration, machine learning, signal processing, and data-driven optimization problems.
- Prepare research manuscripts for publication in high-impact international journals and present research findings at seminars and academic conferences.
- Collaborate with supervisors and researchers on interdisciplinary research projects and contribute to the advancement of computational optimization methods.

- ✦ Research Assistant | Centre for High Performance Computing, CSU 2020-2023
Key Contributions:

- Established the global and Q-linear convergence rate of the proposed algorithm under standard conditions.
- Ensured the sufficient descent property in the search direction of the algorithm.
- Performed large-scale numerical computations using high-performance computing to demonstrate the algorithm's efficacy compared to existing methods.
- Applied the algorithm to solve applications using high-performance computing.

- ✦ Research Fellow | Numerical Optimization Research Group | Bayero University Kano (BUK), Nigeria 2020-2022
Supervisor: Prof A.B. Abubakar | Iterative Algorithms for Solving Large-scale Systems of Nonlinear Equations with Applications
Key Contributions:

- Conceptualizing and drafting the manuscript; demonstrating convergence, analyzing proofs, preparing experiments, and creating tables and figures. View
- Conceptualization and drafting the manuscript; discussing the convergence, analyzing the proofs, and preparing the image recovery experiments.View

Teaching Experience

- ✦ Assistant Lecturer | SLUK Dec 2019-Till date
Responsibilities:

- Delivered tailored tutoring sessions to undergraduate students, focusing on core mathematics courses such as Calculus I, Calculus II, and Algebra.
- Developed and curated detailed lecture materials, including notes and supplementary resources, specifically designed to meet the academic needs of students in Calculus and Algebra.
- Assisted faculty members by grading student assignments, quizzes, and examinations, and provided timely, constructive feedback to help students improve their performance.
- Engaged with students during office hours to provide additional guidance, clarify complex topics, and assist them in preparing for assessments.

- ✦ Assistant Instructor | National Youth Service Corps (NYSC) | Kano University of Science & Technology, Wudil, Nigeria (KUST) Nov 2018-Nov 2019
Responsibilities:

- Assisted in supervising and invigilating students during university examinations.
- Collaborated with faculty members to design and create visually engaging poster presentations for the department's weekly seminar series, highlighting key research topics and fostering academic discussions.
- Supported the department by grading student assignments, quizzes, and examinations.

Publications, Presentations & Workshops

✧ Journal Publications:

1. **Abdullahi, M.**, Sarkinbai, M. S., Abubakar, A. B., Al-Yaari, A., & Siamakani, S. Y. M. (2026). A two-term hybrid method with restart strategy for signal processing and image recovery. *Scientific Reports*.DOI
2. Al-Yaari, A., Deng, Y., & **Abdullahi, M.** (2026). Generative latent optimization with variational regularization for high-quality image synthesis and inverse problems. *International Journal of Computer Mathematics*, 1-26.DOI
3. Sarkinbai, M. S., **Abdullahi, M.**, Abubakar, A. B., & Siamakani, S. Y. M. (2026). A modified restart strategy for a derivative-free projection method with three-term conjugate-gradient direction for monotone equations and application. *Results in Control and Optimization*, 100733.DOI
4. **Abdullahi, M.**, Saleh, M. A., Siamakani, S. Y. M., Almaymuni, A. Z., Abubakar, A. B., Halilu, A. S., & Ibrahim, S. M. (2026). An inertial three-term conjugate gradient projection method with applications. *Journal of Mathematics*,2026(1),5663059DOI
5. **Abdullahi, M.**, Halilu, A. S., & Muhammed, A. M. (2026). Accelerated double step-length method for solving monotone nonlinear equations with convex-constraint and application. *Aims Mathematics*, 1-65. DOI
6. **Abdullahi, M.**, Pan, K., Halilu, A. S., Abubakar, A. B., Tunde, T. A., & Al-Yaari, A. (2026). A new spectral conjugate gradient method for convex constrained monotone nonlinear equations with application to image restoration.*Japan Journal of Industrial and Applied Mathematics*.DOI
7. **Abdullahi, M.**, Al-Yaari, A., Pan, K., Abubakar, A. B., & Halilu, A. S. (2026). An inertial double-step-length-based method for constrained nonlinear equations with applications. *Computational and Applied Mathematics*, 45(8), 340.DOI
8. Al-Yaari, A., **Abdullahi, M.**, & Abdullahi, U. A. (2026). Generative machine learning models for image synthesis: Advances, challenges, and future directions. *Machine Learning with Applications*. 100895.DOI
9. Al-Yaari, A., Deng, Y., **Abdullahi, M.**, & Shakeel, M. (2026). Coevolutionary algorithms in generative adversarial networks for medical image analysis with limited labels. *Physica Scripta* .DOI
10. Sulaiman, A., Abubakar, A. B., **Abdullahi, M.**, Yayangida, S. A., & Masu, S. B. (2026). A New Class of Hybrid Spectral Conjugate Gradient Approach for Unconstrained Optimization and Motion Control Problems. *Bangmod International Journal of Mathematical and Computational Science*, 21-42.DOI
11. **Abdullahi, M.**, Al-Yaari, A., Abubakar, A. B., Halilu, A. S., & Muhammed, A. M. (2025). An effective double-direction method for solving convex-constrained nonlinear equations with image restoration. *Aims Mathematics*, 1-65. DOI
12. **Abdullahi, M.**, Pan, K., Halilu, A. S., & Abubakar, A. B. (2025). An accelerated derivative-free method using Picard-Krasnoselskii iterative process with applications in motion control. *International Journal of Computer Mathematics*, 1-25. DOI
13. **Abdullahi, M.**, Pan, K., Halilu, A. S., Abubakar, A. B., & Zakari, F. I. (2025). Two accelerated double-direction methods for convex-constrained nonlinear equations with applications. *Numerical Algorithms*, 1-28.. DOI
14. **Abdullahi, M.**, Halilu, A. S., Pan, K., & Abubakar, A. B. (2025). Two double-direction methods for convex-constrained nonlinear monotone equations with image recovery of Spherical Parts. *Communications in Nonlinear Science and Numerical Simulation*, 109315. DOI
15. Abbass, G., Chen, H., **Abdullahi, M.**, Halilu, A. S., & Ahmed, M. (2025). Hybrid accelerated double step length approach for convex constrained nonlinear monotone equations with applications. *IMA Journal of Applied Mathematics*, 2025; 90:75–98. DOI
16. **Abdullahi, M.**, Kejia, P., Abubakar, A. B., & Halilu, A. S. (2025). An Inertial Spectral Conjugate Gradient Method for Monotone Nonlinear Equations With Applications. *Mathematical Methods in the Applied Sciences*, 2025; 0:1-16. DOI
17. Ebenezer, N., **Abdullahi, M.**, & Halilu, A. S. (2024). Modification of a conjugate gradient approach for convex-constrained nonlinear monotone equations with applications in signal recovery. *Communication in Nonlinear Science and Numerical Simulation*, 136, 108079. DOI
18. Salihu, S. B., Halilu, A. S.,**Abdullahi, M.**, Ahmed, K., Mehta, P., & Murtala, S. (2024). An improved spectral conjugate gradient projection method for monotone nonlinear equations with application. *Journal of Applied Mathematics and Computing*, 1-37. DOI
19. Abbass, G., Chen, H., **Abdullahi, M.**, & Muhammad, A. B. (2024). An efficient projection algorithm for large-scale system of monotone nonlinear equations with applications in signal recovery. *Journal of Industrial and Management Optimization*, 20(11), 3580-3595. DOI
20. **Abdullahi, M.**, Abubakar, A. B., Sulaiman, A., & Chotpitayasunon, P. (2024). An efficient projection algorithm for solving convex-constrained monotone operator equations and sparse signal reconstruction problems. *Journal of Analysis*, 1-20. DOI

21. Abubakar, A. B., Ibrahim, A. H., **Abdullahi, M.**, Aphane, M & Jiawei, C. (2024). A sufficient descent LS-PRP-BFGS-like method for solving nonlinear monotone equations with application to image restoration. *Numerical Algorithm*, 96, 1423–1464. DOI
22. Abbass, G., Chen, H., **Abdullahi, M.**, Baba, M. A., & Musa, S. (2023). A projection method for solving monotone nonlinear equations with application. *Physica Scripta*, 98(11), 115250. DOI
23. **Abdullahi, M.**, Abubakar, A. B., & Salihu, S. B. (2023). Global convergence via modified self-adaptive approach for solving constrained monotone nonlinear equations with application to signal recovery problems. *RAIRO-Operations Research*, 57(5), 2561-2584. DOI
24. **Abdullahi, M.**, Abubakar, A. B. & Muangchoo, K. (2023). Modified three-term derivative-free projection method for solving nonlinear monotone equations with application. *Numerical Algorithm* 95, 1459–1474. DOI
25. **Abdullahi, M.**, Abubakar, A.B., Feng, Y., & Liu, J. (2023). Comment on: “A derivative-free iterative method for nonlinear monotone equations with convex constraints”. *Numerical Algorithm* 94, 1551–1560. DOI
26. **Abdullahi, M.**, Halilu, A., Awwal, A., & Pakkaranang, N. (2021). On efficient matrix-free method via quasi-Newton approach for solving system of nonlinear equations. *Advances in the Theory of Nonlinear Analysis and its Application*, 5(4), 568-579. DOI

✧ **Conferences Presentation:**

1. **Abdullahi, M.**; An improved conjugate gradient method for solving a system of nonlinear equations. Presented at the 57th Annual Conference of the Mathematical Association of Nigeria (MAN), Kano, Aug, 2021.View
2. **Abdullahi, M.**; An effective matrix-free method via quasi-newton update for solving large-scale system of nonlinear equations. Presented at the 37th Annual National Conference of Nigerian Mathematical Society (NMS), Kano, May 8-11, 2018.View

✧ **Workshops & Webinars:**

1. International Webinar on Intelligence Optimization Techniques and Their Real-Life Applications. Department of Mathematics, Punjabi University, Patiala, India. Feb, 2022.View
Knowledge gain: Gain insight into advanced optimization techniques such as Genetic Algorithms (GA), Particle Swarm Optimization (PSO), and their applications in fields like energy management, logistics, and system design.
2. Numerical and Analytical Techniques in Engineering Problems. Department of Mathematics, Institute of Science and Technology, Tamil Nadu, India. Nov, 2022.View
Knowledge gain: Learn alternative methods for using numerical techniques to solve nonlinear equations, perform integration, and address real-world applications.
3. Nonlinear Operator Theory, Convex Optimization, and Partial Differential Equations. Institute of Mathematics, African University of Science and Technology, Abuja. Online 2022.
Knowledge gain: Acquired in-depth knowledge of advanced mathematical concepts, including nonlinear operator theory, convex optimization techniques, and their applications in solving partial differential equations.
4. Numerical and Analytical Techniques in Engineering Problems. Department of Mathematics, Institute of Science and Technology, Tamil Nadu, India. Nov, 2020.View
Knowledge gain: Learn the principles of various numerical techniques for solving nonlinear equations, performing integration, and solving differential equations using the Runge-Kutta methods.
5. International Forum on The Current Advances in Partial Least Squares Structural Equation Modeling & Methodological Matters. Emerald Publishing. Jun, 2020.View
Knowledge gain: A deep understanding of the latest methodological advancements in PLS-SEM, along with the ability to apply cutting-edge techniques to complex models effectively, enhances both my research rigor and practical application skills.

Volunteer, & Leadership Experience

✧ Reviewer Franklin Open, Elsevier.View	2024-Till date
✧ Reviewer Systems and Soft Computing, Elsevier.View	2024-Till date
✧ Reviewer Applied Soft Computing, Elsevier.View	2024-Till date
✧ Reviewer Scientific African, Elsevier. View	2024-Till date
✧ Class Representative Department of Mathematics SLUK Responsibilities:	2015-2018
• Acted as the intermediary between students, faculty, and school management on issues related to students, including assignment submissions, addressing exam or result complaints, and organizing academic events.	

✠ Community Tutorial Coordinator | GRA Hadejia
Responsibilities:

Nov 2015-Till date

- Managed a team of volunteers for a weekly tutorial in Maths, Physics, Biology, and Chemistry at Government Girls Science Secondary School, Hadejia.
- Implemented a new volunteer training program, increasing volunteer retention by 30%.

Skills

- **Software Skills:** Matlab, Maple, and Python
- **Soft Skills:** Leadership, Public Speaking, Time Management

Scholarships, Grants & Awards

✠ Chinese Government Council Scholarship.View	2020
✠ Bilateral Education Agreement Scholarship Award Federal Scholarship Board, Abuja, NigeriaView	2020
✠ HOD's Prize for Best Graduating Student SLUK.View	2018
✠ Jigawa State Government Scholarship: Two consecutive payments of ₦40,000	2016 & 2018

Extracurricular Activities

Assistant Secretary | Active Citizens Club SLUK Branch

Nov 2015-Aug 2018

Activities:

- Bring people together to address common issues, plan community activities, and develop local initiatives.
- Informed oneself and others about important issues, policies, and events that have an impact on the community and beyond.

References

Available upon request.